Shelley S. Vaseegar

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ABOUT ME

I believe AI's greatest value is in making human work easier, more meaningful, and more impactful. That conviction drives how I approach every problem: not just building models that perform well, but designing systems that people can actually use and trust. CS graduate with research-grade deep learning work published at ICISC 2026, enterprise systems debugging across 50+ customer instances at ServiceNow, and end-to-end automation delivery at SPI Edge. Fast learner, 9.45 GPA, consistent track record of exceeding expectations in every new context I step into.

SKILLS

AI/ML: PyTorch, TensorFlow, Scikit-learn, Hugging Face, Swin Transformer, ViT, ResNet, VGG, Grad-CAM++, Integrated Gradients
Generative AI: RAG pipelines, LLM integration, prompt engineering, vector embeddings, ChromaDB, BioBERT
Languages & Data: Python, SQL, C++, JavaScript; PostgreSQL, MySQL; EDA, feature engineering, hypothesis testing
Backend & APIs: FastAPI, Flask, SQLAlchemy, REST APIs, React, OpenCV, n8n
Systems & Tools: Linux (SSH, Splunk, JVM profiling, Eclipse MAT), Git, Agile/Scrum

WORK EXPERIENCE

ServiceNow | Associate Technical Support Engineer Intern — *Jan 2026 – Jun 2026*

- **OOM & JVM analysis:** Diagnosed Out-of-Memory failures on application nodes by analyzing heap dumps, thread dumps, and GC logs using Eclipse MAT and Splunk on live Linux environments; produced customer RCA documents across multiple root cause categories
- **Database investigations:** Developed 12 validated shard-aware SQL query patterns (MySQL + PostgreSQL) used across 50+ customer instances to trace sysevent backlogs, replication lag, and connection pool saturation
- **Internal tool:** Built Chrome/Edge extension (MV3, Shadow DOM, Python/internal LLM API) that auto-drafts customer-facing case notes, shipped to full TSE team, saving 5 to 10 minutes per session
- **Automation tooling:** Delivered Teams message extractor, SLA deadline reminder workflow, and React-based troubleshooting tool using Python and internal APIs, reducing debugging time by ~1 hour/day
- **Case ownership:** Investigated and closed 60+ cases across incident types under enterprise SLAs
- **Product defect identification:** Traced an unusual script-level bug during a live customer investigation, isolated root cause, and handed off a full RCA to the development team for resolution

SPI Edge | Business Analyst & Automation Intern — *Jun 2025 – Jul 2025*

- **Ticket triaging (Zoho Desk + n8n + RAG):** Built classification, priority assignment, routing, and LLM-drafted response pipeline via vector store + n8n; cut manual routing by 60% and FRT by 2.5 hours
- **Asset tracking (Google Forms + Telegram Bot + Notion):** Built 3-layer request/approval/tracking system with inline Telegram action buttons; reduced equipment mismanagement by 40%
- **Self-service chatbot (Voiceflow):** Deployed FAQ + lead capture bot with LLM integration; cut API costs by ₹20,000+/month and improved task completion rates

PROJECT EXPERIENCE

HSwin-OTD: Explainable Multi-Class Intraocular Tumor Classification | *PyTorch, Swin Transformer, ChromaDB, BioBERT, Grad-CAM++*
Research paper accepted at ICISC 2026, co-authored with affiliates from SRMIST and the University of Oxford.

- Designed 4-stage hierarchical Swin Transformer for multi-class tumor classification on 2,031 UWF retinal images with severe class imbalance, conservative offline oversampling with stratified 80/10/10 split
- Achieved 97.43% accuracy (0.97 F1-score) with AdamW + cosine annealing, outperformed ViT (+9%), VGG16 (+15%), ResNet50 (+18%)
- Multi-layer XAI framework: Grad-CAM++ + Integrated Gradients across all 4 Swin stages for clinically interpretable predictions
- RAG pipeline: 278 research papers chunked into 3,342 segments, embedded using BioBERT, stored in ChromaDB; generates literature-grounded diagnostic explanations

Drowsiness Detection System in Commercial Vehicles | *OpenCV, Flask, Python, Linux*

- Real-time fatigue detection pipeline using EAR, MAR, PERCLOS with hybrid Haar Cascades + HOG-SVM face detection; robust across varying lighting conditions and facial orientations.
- Implemented multi-threshold alert system triggering graduated warnings based on drowsiness severity; end-to-end pipeline from frame capture to alert with minimal latency
- Finalist in a national hackathon. Presented solution to domain experts, incorporated feedback, and published patent (Application No: 202541037313) for the scalable approach

Query Optimization & Backend Performance System | *FastAPI, SQLAlchemy, PostgreSQL*

- Built REST API backend to execute, monitor, and optimize SQL queries; integrated EXPLAIN ANALYZE for bottleneck detection, improving query efficiency by 25 to 40%
- Rule-based optimization engine providing indexing and filtering insights; reduced slow queries by ~30% across 50K+ row datasets with persistent query storage for comparative analysis

CroPro: Crop Yield Optimization Web Application | *TensorFlow, PyTorch, Scikit-learn, Flask*

- End-to-end agri-tech platform with ML models for crop, disease, and irrigation recommendations using weather and geospatial data; conducted user interviews with farmers to validate feature gaps

EDUCATION

SRM Institute of Science and Technology, Kattankulathur | B.Tech in Computer Science and Engineering
GPA: 9.4 / 10 — *May 2026*
Lalaji Memorial Omega International School | XII Board CBSE | 89.83% — *Jun 2022*
Lalaji Memorial Omega International School | X Board CBSE | 90.83% — *Mar 2020*